

Akash Karthik

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EDUCATION

Bachelor of Science, Computer Science

Expected May 2027

San Jose State University, San Jose, CA

Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Discrete Math, Computer Systems, Intro to Circuits

TECHNICAL SKILLS

Languages: TypeScript/JavaScript, Python, SQL, C/C++, Java

Frontend: React, HTML/CSS/JavaScript, client-side state management, responsive UI

Backend & Data: FastAPI, Flask, Node.js/Express, REST/JSON, WebSockets, real-time streaming, SQLite, SQL Server, schema design

Performance & Delivery: Real-time media pipelines, OpenCV, FPS and latency instrumentation, pytest, Jest, GitHub Actions, Docker, Git/GitHub

EXPERIENCE

Software Lead

Jun 2026 - Present

Spartan Racing, Formula SAE | San Jose, CA

- Lead software planning for Python tooling, validation, and shared testing workflows across five project teams
- Standardize requirements, test evidence, and technical reviews spanning dashboard, simulation, and vehicle software

Software Engineer Intern

Feb 2026 - Present

Argus Defense | San Jose, CA

- Developed and validated flight-control software in C, using structured telemetry logs to reproduce issues and evaluate changes
- Integrated visual-inertial odometry and iterated configuration through repeatable tests and documented results

Software Development Intern

Sep 2024 - Oct 2024

Zip Intelligence | Tracy, CA

- Took ownership of a feature from ticket through verified production release, shipping via pull requests with setup and change documentation
- Implemented frontend validation and clearer error states, then paired with engineers to harden REST endpoints for status codes and edge cases
- Added SQL Server update scripts and structured logs to speed reproduction and issue triage

PROJECT EXPERIENCE

Tickr — AI-Coached Paper-Trading Game

2025

TypeScript/JavaScript, React, LLM integration, responsive UI

- Built registration, sign-in, game, and history flows end-to-end for a scenario-driven trading game with an LLM-backed coach
- Implemented quest and prompt logic, client-side state management, and a mobile-responsive interface

Posture Detection System — Real-Time Media Pipeline

2025

Python, OpenCV, real-time video processing, performance instrumentation

- Built an on-device pipeline that captures and processes live video frames with temporal smoothing, debounce logic, and configurable alert thresholds
- Instrumented frame rate and end-to-end latency, recorded sessions, and exported labeled frames and CSV metrics to tune the pipeline against real usage

SR-Wireless-CAN — Real-Time Telemetry Platform

2026

Spartan Racing, Formula SAE | San Jose, CA | Backend lead

Python, FastAPI, WebSockets, React, SQLite, Docker Compose

- Led backend development for a real-time telemetry platform, building a FastAPI and WebSocket service that streams live data to a React dashboard with SQLite persistence
- Designed the schema and merge logic for signal definitions with conflict resolution, and a bounded search window keeping large catalogs responsive